

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

1) Network Scenario: We consider a general multi-beam LEO satellite communication system under NGSO–GSO coexistence. Since the LEO downlink and the GSO system may operate in the same frequency band, the GSO ground station must be protected from excessive co-channel interference. Hereafter, this ground station is referred to as the GS. A OneWeb-like constellation with 16 fixed spot beams per satellite, as shown in Fig. 1, is used as a representative simulation setting, but the proposed framework can be applied to other LEO systems with predictable beam footprints and coverage overlap.

Let $\mathcal{S}(t)$ denote the set of visible LEO satellites at time slot t , and let $\mathcal{B}_s$ denote the set of beams on satellite s . In the adopted simulation setting, each satellite has 16 beams, while the formulation can be extended to systems with different beam numbers.

Each satellite employs a fixed multi-beam layout. For beam b on satellite s , the ground footprint at time slot t is denoted by $\mathcal{F}_{s,b}(t)$. If user u lies in $\mathcal{F}_{s,b}(t)$, beam b on satellite s is regarded as a feasible serving candidate for that user. When multiple satellite–beam pairs cover the same ground area, the overlap provides possible serving alternatives through user association and resource coordination.

When a visible LEO satellite appears close to the GS antenna boresight or forms a near-inline geometry with the desired GSO satellite direction, its downlink transmission may produce a high aggregate EPFD at the GS. The LEO satellite that contributes the largest EPFD risk at time slot t is defined as the *critical satellite*, denoted by $s^C(t)$.

We further consider the along-track north and south neighboring satellites of the critical satellite as candidate helper satellites for subsequent service transfer and relaying, denoted by $s^N(t)$ and $s^S(t)$, respectively, as shown in Fig. 2. Since LEO satellite motion and beam footprints are predictable, the coverage overlap between critical beams and helper beams can be estimated in advance, enabling helper beams to relay users affected by critical-beam shutdown.

2) User Association and Service Model: Let $\mathcal{U}(t)$ denote the set of terrestrial users considered at time slot t , where each user is covered by at least one visible LEO beam. Each user can be associated with at most one satellite–beam pair in the same time slot. The user–beam association indicator $x_{u,s,b}(t)$ is defined as

$$x_{u,s,b}(t) = \begin{cases} 1, & \text{if user } u \text{ is served by beam } b \text{ of satellite } s, \\ 0, & \text{otherwise,} \end{cases}$$

for all $u \in \mathcal{U}(t)$, $s \in \mathcal{S}(t)$, and $b \in \mathcal{B}_s$.

For beam b on satellite s , the number of users served by this beam is

$$N_{s,b}(t) = \sum_{u \in \mathcal{U}(t)} x_{u,s,b}(t). \quad (1)$$

This value is used in the TDMA-based rate model, where users served by the same beam share the available downlink resource.

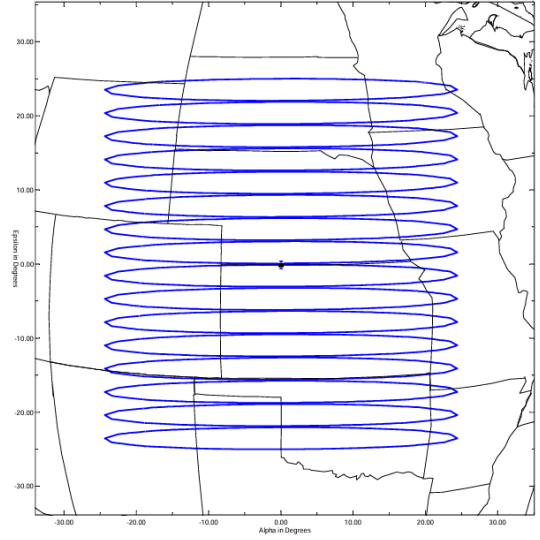


Fig. 1. Ground footprints of the OneWeb Ku-band 16-beams

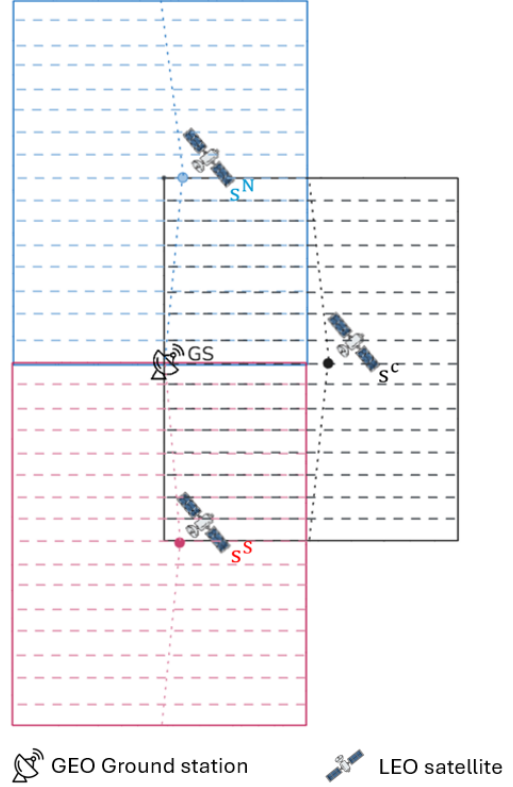


Fig. 2. Network scenario with the protected GSO ground station GS , critical satellite s^C , and north/south helper satellites s^N and s^S .

An association is feasible only when user u lies within the footprint $\mathcal{F}_{s,b}(t)$ of beam b on satellite s . Therefore, association, reassociation, and relay decisions are made only among satellite–beam pairs that can actually cover the user. In overlapping coverage areas, a user may be covered by multiple satellite–beam pairs. Under normal operation, these overlapping beams can coordinate the service demand through user association, time-domain scheduling, or spatial separation, while each user

is still associated with only one serving beam in each time slot.

When a critical beam is shut down to satisfy the GS EPFD constraint, the users originally served by this beam can no longer be supported by the critical satellite. If these users are also covered by nearby helper beams, they can be reassocated to feasible helper relay beams. Since the helper relay beams use a compatible downlink service configuration, such as the same service band and polarization setting, the reassocation can be described within the same service model.

3) Beam Power Model: We adopt a default beam-power operation model. For beam b on satellite s , the transmit power at time slot t is denoted by $P_{s,b}(t)$. Under normal operation, when LEO satellites are far from the GS and the EPFD violation risk is low, each active beam operates with its preconfigured default transmit power, denoted by $P_{s,b}^{\text{def}}$.

Different beams are not required to use the same default transmit power. The default power of each beam is preconfigured based on its link condition, so that users in different beam footprints can receive a similar service level under normal operation. Therefore,

$$P_{s,b}(t) = P_{s,b}^{\text{def}}, \quad \forall s \in \mathcal{S}(t), \forall b \in \mathcal{B}_s.$$

This default beam-power profile serves as the normal operating state. If the GS EPFD constraint cannot be satisfied under this state, some beam powers may be adjusted or shut down.

Each beam has a maximum transmit power $P_{s,b}^{\text{max}}$, and

$$0 \leq P_{s,b}(t) \leq P_{s,b}^{\text{max}}, \quad \forall s \in \mathcal{S}(t), \forall b \in \mathcal{B}_s. \quad (2)$$

Each satellite also has a total transmit power budget P_s^{max} :

$$\sum_{b \in \mathcal{B}_s} P_{s,b}(t) \leq P_s^{\text{max}}, \quad \forall s \in \mathcal{S}(t). \quad (3)$$

When the critical satellite $s^C(t)$ creates an aggregate EPFD violation risk at the GS, some critical beams may be shut down. Let $\mathcal{B}_C^{\text{off}}(t)$ denote the set of beams shut down on the critical satellite, also referred to as closed beams. Then

$$P_{s^C,b}(t) = 0, \quad \forall b \in \mathcal{B}_C^{\text{off}}(t).$$

The remaining active beams on the critical satellite are referred to as safe beams and are denoted by $\mathcal{B}_C^{\text{safe}}(t)$, where

$$\mathcal{B}_C^{\text{safe}}(t) = \mathcal{B}_{s^C} \setminus \mathcal{B}_C^{\text{off}}(t).$$

These safe beams remain active with their default transmit powers:

$$P_{s^C,b}(t) = P_{s^C,b}^{\text{def}}, \quad \forall b \in \mathcal{B}_C^{\text{safe}}(t).$$

Hence, the transmit power of a beam on the critical satellite can be written as

$$P_{s^C,b}(t) = \begin{cases} 0, & b \in \mathcal{B}_C^{\text{off}}(t), \\ P_{s^C,b}^{\text{def}}, & b \in \mathcal{B}_C^{\text{safe}}(t). \end{cases}$$

For helper satellites, beams also operate with their default transmit powers under normal operation. If users affected by critical-beam shutdown are reassocated to feasible helper beams, the transmit power of those helper beams may be

adjusted above their default values, subject to the beam-level maximum power and the total satellite power budget (3):

$$P_{s^h,b}(t) \leq P_{s^h,b}^{\text{max}}, \quad s^h \in \{s^N(t), s^S(t)\}.$$

Throughout this model, each user is still served by only one satellite-beam pair, and the service rate is determined by the selected serving beam and the users sharing that beam.

4) SINR Model: The downlink SINR of a user is modeled using a commonly adopted wireless and satellite communication formulation, where the received signal, co-channel interference, and receiver noise are considered [15]. Let $k_{s,b}$ denote the downlink channel used by beam b on satellite s . Only beams using the same downlink channel as the serving beam are included as co-channel interferers.

If user u is served by beam b on satellite s , its downlink SINR at time slot t is

$$\text{SINR}_u(t) = \frac{S_u(t)}{I_u^{\text{intra}}(t) + I_u^{\text{inter}}(t) + N_u(t)}. \quad (4)$$

where $S_u(t)$ is the desired signal power, $I_u^{\text{intra}}(t)$ is the intra-satellite co-channel interference, $I_u^{\text{inter}}(t)$ is the inter-satellite co-channel interference, and $N_u(t)$ is the receiver noise power.

Desired Signal Power: The desired signal power follows the standard satellite link-budget relation, and the propagation loss components are modeled according to ITU-R recommendations [9], [10]. For user u served by beam b on satellite s ,

$$S_u(t) = \frac{P_{s,b}(t) G_{s,b \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s,u}(t)}.$$

where $P_{s,b}(t)$ is the transmit power of the serving beam, $G_{s,b \rightarrow u}^{\text{tx}}(t)$ is the transmit gain of beam b toward user u , G_u^{rx} is the user receive gain, and $L_{s,u}(t)$ is the total propagation loss between satellite s and user u .

The total propagation loss includes free-space path loss and atmospheric gaseous loss:

$$L_{s,u}(t) = L_{s,u}^{\text{fs}}(t) L_{s,u}^{\text{gas}}(t),$$

where $L_{s,u}^{\text{fs}}(t)$ is the free-space path loss and $L_{s,u}^{\text{gas}}(t)$ is the atmospheric gaseous loss.

The free-space path loss follows the standard free-space attenuation model in ITU-R P.525 [9]:

$$L_{s,u}^{\text{fs}}(t) = \left(\frac{4\pi d_{s,u}(t)}{\lambda} \right)^2,$$

where $d_{s,u}(t)$ is the distance between satellite s and user u , and λ is the carrier wavelength.

Intra-Satellite Interference: The intra-satellite interference comes from co-channel beams on the same satellite. Let $\mathcal{B}_s^{\text{co}}(b)$ denote the set of beams on satellite s that reuse the same channel as the serving beam b , excluding beam b itself. The intra-satellite interference is given by

$$I_u^{\text{intra}}(t) = \sum_{b' \in \mathcal{B}_s^{\text{co}}(b)} \frac{P_{s,b'}(t) G_{s,b' \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s,u}(t)}.$$

Inter-Satellite Interference: The inter-satellite interference comes from co-channel beams on other visible satellites. Let

$\mathcal{B}_{s'}^{\text{co}}(b)$ denote the set of beams on satellite s' that reuse the same channel as the serving beam b ; The inter-satellite interference is given by

$$I_u^{\text{inter}}(t) = \sum_{\substack{s' \in \mathcal{S}(t) \\ s' \neq s}} \sum_{b' \in \mathcal{B}_{s'}^{\text{co}}(b)} \frac{P_{s',b'}(t) G_{s',b' \rightarrow u}^{\text{tx}}(t) G_u^{\text{rx}}}{L_{s',u}(t)}.$$

where $G_{s',b' \rightarrow u}^{\text{tx}}(t)$ is the transmit gain of the interfering beam toward user u , and $L_{s',u}(t)$ is the total propagation loss between satellite s' and user u .

With this commonly used downlink SINR model, the user SINR changes when the serving beam, beam transmit power, propagation loss, or co-channel interference changes.

5) Capacity and User Satisfaction Model: After obtaining the user SINR, the instantaneous link capacity is modeled by the Shannon capacity formula, which is commonly used in wireless and satellite communication systems [16], [17]. Let B_u denote the effective bandwidth allocated to user u . The instantaneous capacity of user u at time slot t is

$$C_u(t) = B_u \log_2(1 + \text{SINR}_u(t)).$$

When multiple users are served by the same beam, they share the downlink resource of that beam. Following the time-division resource sharing model commonly used in satellite downlink systems [17], [18], the beam resource is assumed to be equally divided among the users served by the same beam. The actual service rate of user u is therefore

$$R_u(t) = \frac{C_u(t)}{N_{s,b}(t)}. \quad (5)$$

where $N_{s,b}(t)$ is defined in (1). This expression represents an equal-time sharing model, where users served by the same beam are allocated the same fraction of the beam resource in each time slot.

Let $D_u(t)$ denote the demand rate of user u at time slot t . The satisfaction of user u is defined as

$$\text{sat}_u(t) = \min\left(\frac{R_u(t)}{D_u(t)}, 1\right). \quad (6)$$

where $\text{sat}_u(t) \in [0, 1]$. If $\text{sat}_u(t) = 1$, the demand of user u is fully met; if $0 < \text{sat}_u(t) < 1$, user u receives partial service; if $\text{sat}_u(t) = 0$, user u receives no effective service in the time slot.

This satisfaction metric represents the fraction of user demand that is satisfied in each time slot.

6) Beam-Level EPFD Model: The EPFD calculation follows the ITU Radio Regulations and ITU-R S.1503 framework for NGSO–GSO coexistence [5], [6]. Since different beams on the same LEO satellite have different pointing directions, antenna gains, and transmit powers toward the GS, the EPFD contribution is evaluated at the beam level and summed in the linear domain.

For EPFD calculation, let $P_{s,b}^{\text{ref}}(t)$ denote the radio-frequency (RF) transmit power of beam b on satellite s within the ITU reference bandwidth B_{ref} specified for the applicable EPFD limit [7], [8]. If $P_{s,b}(t)$ is defined over a wider beam bandwidth, it is converted to the corresponding power within B_{ref} before

applying the EPFD formula. The beam-level EPFD contribution toward the GS is given by

$$\text{EPFD}_{s,b}(t) = \frac{P_{s,b}^{\text{ref}}(t) G_{s,b \rightarrow g}^{\text{tx}}(t)}{4\pi d_{s,g}^2(t)} \cdot \frac{G_g^{\text{rx}}(\theta_{g \rightarrow s}(t))}{G_{g,\text{max}}^{\text{rx}}},$$

where $G_{s,b \rightarrow g}^{\text{tx}}(t)$ is the transmit antenna gain of beam b on satellite s in the direction of the GS, $d_{s,g}(t)$ is the distance between satellite s and the GS, $G_g^{\text{rx}}(\theta_{g \rightarrow s}(t))$ is the receive antenna gain of the reference GS antenna in the direction of satellite s , $G_{g,\text{max}}^{\text{rx}}$ is the maximum receive antenna gain of the reference GS antenna, and $\theta_{g \rightarrow s}(t)$ is the off-axis angle between the GS antenna boresight and the direction of satellite s .

The transmit antenna gain $G_{s,b \rightarrow g}^{\text{tx}}(t)$ is evaluated from the beam antenna pattern according to the off-axis angle between the boresight of beam b and the direction of the GS. Since each beam has a different pointing direction, the EPFD contribution of each beam is computed individually.

The aggregate EPFD at the GS is obtained by summing the beam-level EPFD contributions from all beams of all visible LEO satellites:

$$\text{EPFD}_{\text{agg}}(t) = \sum_{s \in \mathcal{S}(t)} \sum_{b \in \mathcal{B}_s} \text{EPFD}_{s,b}(t).$$

The resulting aggregate EPFD is compared with the applicable ITU EPFD limit in the problem formulation.

B. Problem Formulation

Based on the system model above, we formulate the user association and beam power control problem to maximize the total user satisfaction while satisfying association, power, and EPFD constraints. The optimization variables are the user–beam association indicators $\mathbf{x}(t) = \{x_{u,s,b}(t)\}$ and the beam transmit powers $\mathbf{P}(t) = \{P_{s,b}(t)\}$.

Objective function:

$$\max_{\mathbf{x}(t), \mathbf{P}(t)} \sum_{u \in \mathcal{U}(t)} \text{sat}_u(t). \quad (7)$$

Constraints:

$$\sum_{s \in \mathcal{S}(t)} \sum_{b \in \mathcal{B}_s} x_{u,s,b}(t) \leq 1, \quad \forall u \in \mathcal{U}(t), \quad (8)$$

$$x_{u,s,b}(t) \in \{0, 1\}, \quad \forall u \in \mathcal{U}(t), \quad \forall s \in \mathcal{S}(t), \quad \forall b \in \mathcal{B}_s, \quad (9)$$

$$0 \leq P_{s,b}(t) \leq P_{s,b}^{\text{max}}, \quad \forall s \in \mathcal{S}(t), \quad \forall b \in \mathcal{B}_s. \quad (10)$$

$$\sum_{b \in \mathcal{B}_s} P_{s,b}(t) \leq P_s^{\text{max}}, \quad \forall s \in \mathcal{S}(t). \quad (11)$$

$$\text{EPFD}_{\text{agg}}(t) \leq \text{EPFD}_{\text{lim}}. \quad (12)$$

Constraint (8) ensures that each user is associated with at most one satellite–beam pair in each time slot.

Constraint (9) defines the binary association decision, where $x_{u,s,b}(t)$ can only be 0 or 1 and therefore cannot take a negative value.

Constraint (10) limits the transmit power of each beam, including the possibility of beam shutdown when $P_{s,b}(t) = 0$.

Constraint (11) enforces the total transmit power budget of each satellite.

Constraint (12) ensures that the aggregate EPFD at the GS does not exceed the applicable ITU limit.

The objective in (7) maximizes the aggregate satisfaction of all users in the considered time slot. Although the objective is written as a summation of user satisfaction, each satisfaction value depends on the selected serving beam, beam loading, SINR, and service rate. In addition, beam power affects both the user-side downlink performance and the GS-side aggregate EPFD. Therefore, the association and power decisions are strongly coupled.

In critical GS geometries, reducing or shutting down beam power can lower the aggregate EPFD, but it may also reduce the service rate of users originally served by the affected beams. These users can be reassigned to feasible helper beams, but doing so changes the helper beam loading and may further affect the service rates of other users.

Because of these coupled association, power, SINR, service-rate, and EPFD relationships, the problem becomes a mixed-integer nonlinear optimization problem. Solving the global optimum in every time slot is computationally difficult. Therefore, the next chapter develops a staged heuristic framework to obtain feasible association and power decisions during critical periods.

IV. METHOD

A. Method Overview

Chapter III formulates the problem as maximizing total user satisfaction subject to user association, beam power, satellite total power, and EPFD constraints. However, the problem involves both discrete user-beam association decisions and continuous beam power configuration, and association, SINR, TDMA sharing, satisfaction, and aggregate EPFD are tightly coupled. Solving for a global optimum is therefore computationally demanding.

To reduce online decision complexity, we propose Safe-Beam Assisted Power-Release Relay (SAPR-R). SAPR-R exploits the predictability of the LEO constellation, protected GS location, GSO direction, and fixed beam layout to pre-build beam-level relation records for all time slots in the observation window. For each time slot, the system records the critical satellite, north/south helper satellites, beams on the critical satellite that must be shut down, safe beams that remain active at baseline power, and beam-pair mappings between critical and helper beams. This stage depends only on satellite geometry, beam footprints, and EPFD calculation—not on user distribution—and can be completed before online user reassociation.

During online operation, the system reads the beam relation record for the current time slot and performs user-aware relay decisions based on user locations and associations. First, SAPR-R applies Safe-Beam Assisted Helper Relief (SAHR) so that critical safe beams attempt to take over users on helper release beams. If reassociation does not reduce the satisfaction of the transferred user or of existing users on the safe beam, the transfer is accepted and the power released from the helper

release beam is returned to the helper satellite's available power pool. Next, the helper satellite allocates the power pool to helper relay beams according to relay demand, increasing capacity for subsequent service to closed-beam users. Finally, SAPR-R applies Beam-Pair Local Relay (BPLR): within each closed-beam/helper relay-beam pair, critical closed-beam users are ranked and allocated sequentially according to relay link quality.

In summary, the proposed method consists of the following steps:

- **Time-slot beam relation record:** Pre-record critical/helper satellites, closed/safe beams, and beam-pair mappings for each time slot.
- **Safe-Beam Assisted Helper Relief (SAHR):** Use critical safe beams to take over selected helper users and release helper satellite power without lowering satisfaction.
- **Helper relay beam boosting:** Allocate the available power pool to helper relay beams according to relay demand.
- **Beam-Pair Local Relay (BPLR):** Within each beam pair, allocate relay capacity to closed-beam users according to relay link quality.

B. Time-Slot Beam Relation Record

For each time slot t , we construct a beam relation record $\mathcal{R}_B(t)$. Figure 3 illustrates the time-slot beam status considered in this work, including the:

$$\mathcal{R}_B(t) = (s^c(t), s^N(t), s^S(t), \mathcal{B}_c^{\text{off}}(t), \mathcal{B}_c^{\text{safe}}(t), \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t), \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)).$$

This record stores the beam-level information required by SAHR and BPLR, where $s^c(t)$, $s^N(t)$, and $s^S(t)$ are the critical, north helper, and south helper satellites, respectively. $\mathcal{B}_c^{\text{off}}(t)$ is the set of beams on the critical satellite shut down due to the EPFD constraint; $\mathcal{B}_c^{\text{safe}}(t)$ is the set of safe beams that remain on at baseline power. $\mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$ and $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$ denote the safe-beam-to-helper-release-beam mapping and the closed-beam-to-helper-relay-beam mapping, respectively.

1) Critical and Helper Satellite Identification: In each time slot t , the system identifies the critical satellite that poses the dominant EPFD risk to the protected GS from LEO-GS geometry:

$$s^c(t).$$

From simulation, EPFD violations are dominated by high-risk beams on the critical satellite; we therefore apply shutdown only to beams on $s^c(t)$, not to helper or other visible satellites. The along-track north and south neighbors of the critical satellite are selected as helpers:

$$s^N(t), \quad s^S(t).$$

2) EPFD-Driven Beam Status Determination: For each critical beam $b \in \mathcal{B}_{s^c}$, define the EPFD risk score

$$q_{c,b}(t) = \text{EPFD}_{s^c,b}(t).$$

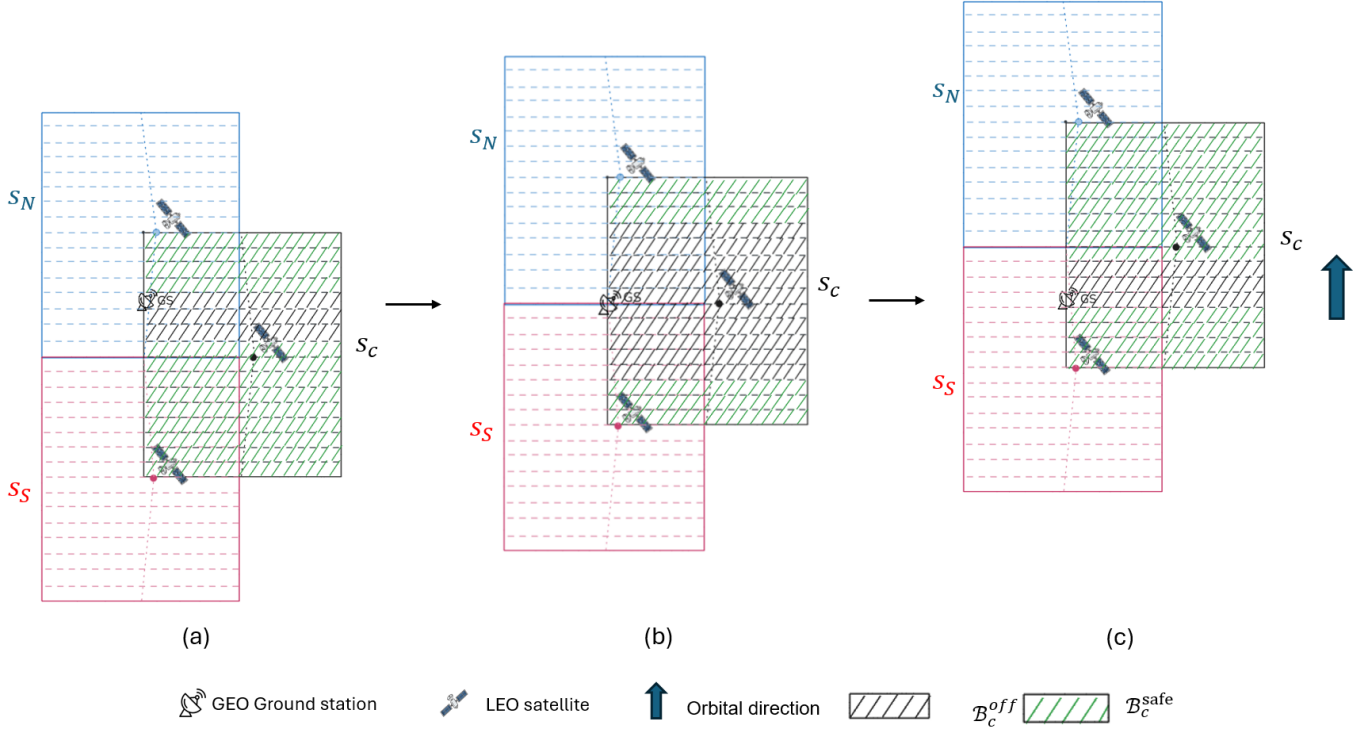


Fig. 3. Three time slots as the along-track LEO satellites (s^N , s^c , and s^S) pass the protected GSO ground station. In region S_C on the critical satellite, black-hatched beams are $\mathcal{B}_c^{\text{off}}(t)$ (closed beams) and green-hatched beams are $\mathcal{B}_c^{\text{safe}}(t)$ (safe beams). Panels (a)–(c) show earlier to later time slots.

Critical beams are sorted in descending order of $q_{c,b}(t)$, and a greedy shutdown rule is applied: at each step, shut down the beam with the highest current EPFD risk score until

$$\text{EPFD}_{\text{agg}}(t) \leq \text{EPFD}_{\text{lim}}.$$

Shut-down beams form

$$\mathcal{B}_c^{\text{off}}(t),$$

and the remaining beams form the safe set:

$$\mathcal{B}_c^{\text{safe}}(t) = \mathcal{B}_{s^c} \setminus \mathcal{B}_c^{\text{off}}(t).$$

Beam status is determined solely from beam-level EPFD contributions and the regulatory limit, without user distribution, and can be precomputed before online reassociation.

3) Beam-Pair Mapping Construction: Given $\mathcal{B}_c^{\text{off}}(t)$ and $\mathcal{B}_c^{\text{safe}}(t)$, beam-pair mappings are built from footprint overlap. For SAHR,

$$\begin{aligned} \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t) = \{ & (b_c^{\text{safe}}, b_h^{\text{rel}}) : \\ & \mathcal{F}_{s^c, b_c^{\text{safe}}}(t) \cap \mathcal{F}_{s_h, b_h^{\text{rel}}}(t) \neq \emptyset, \\ & s_h \in \{s^N(t), s^S(t)\} \}, \end{aligned}$$

which identifies which helper release-beam users a critical safe beam can take over.

For BPLR,

$$\begin{aligned} \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t) = \{ & (b_c^{\text{off}}, b_h^{\text{relay}}) : \\ & \mathcal{F}_{s^c, b_c^{\text{off}}}(t) \cap \mathcal{F}_{s_h, b_h^{\text{relay}}}(t) \neq \emptyset \}, \end{aligned}$$

which identifies which helper relay beams can serve critical closed-beam users.

Figure 4 visualizes the two beam-pair mappings stored in $\mathcal{R}_B(t)$: $\mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$ for SAHR and $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$ for BPLR.

Under the Walker-star constellation and fixed beam layout, the critical satellite and north/south helpers exhibit near one-to-one beam overlap: north-half beams on the critical satellite overlap mainly with south-half beams on the north helper, and south-half critical beams with north-half beams on the south helper. Relay can therefore be executed locally within beam pairs without global matching.

Helper beams are split into release beams (to free helper power) and relay beams (to serve critical closed-beam users). We assume no overlap within the same time slot:

$$\mathcal{B}_h^{\text{rel}}(t) \cap \mathcal{B}_h^{\text{relay}}(t) = \emptyset.$$

C. Safe-Beam Assisted Helper Relief

After loading $\mathcal{R}_B(t)$, the online procedure starts with SAHR. Figure 5 illustrates the SAHR process. The goal is not to serve critical closed-beam users directly, but to use critical safe beams to take over helper users so that helper satellites release power for subsequent relay boosting.

For each

$$(b_c^{\text{safe}}, b_h^{\text{rel}}) \in \mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t),$$

we identify helper users currently served by b_h^{rel} that also lie in the footprint of b_c^{safe} :

$$\begin{aligned} \mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, b_h^{\text{rel}}, t) = \\ \{ v : v \in \mathcal{U}_{s_h, b_h^{\text{rel}}}(t), v \in \mathcal{F}_{s^c, b_c^{\text{safe}}}(t) \}. \end{aligned}$$

Beam-pair mapping visualization for a given time slot t

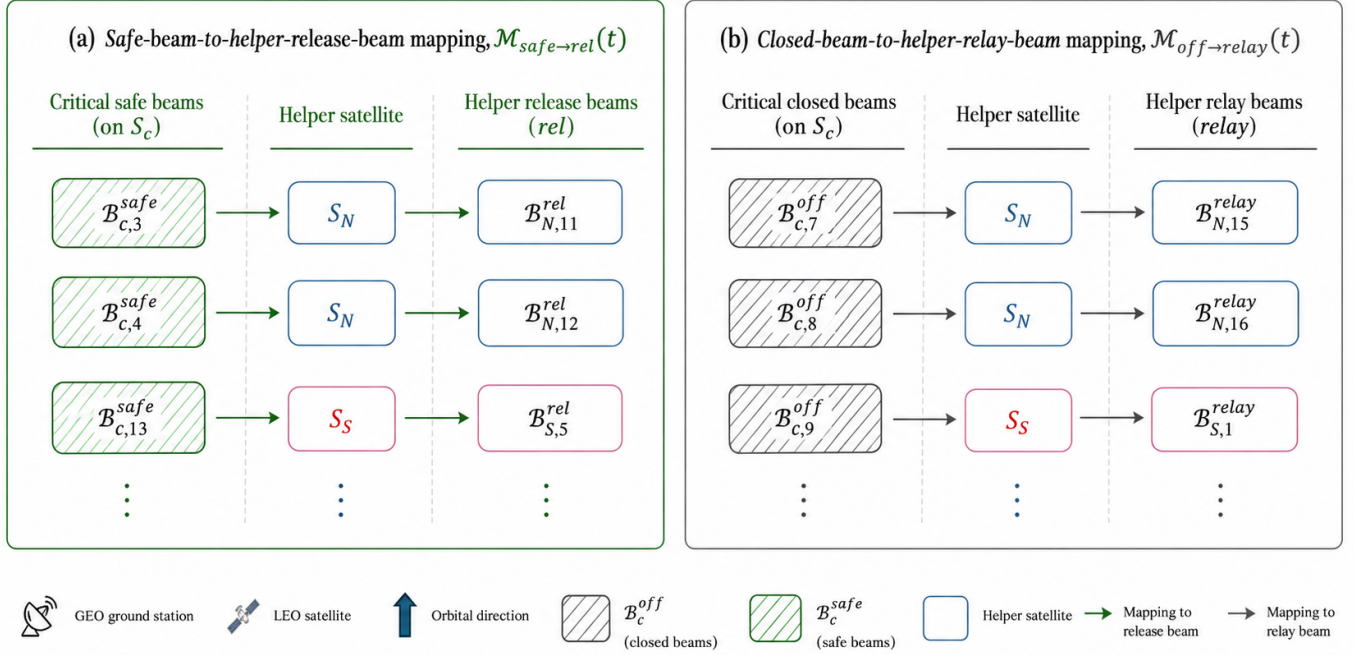


Fig. 4. Beam-pair mapping tables at time slot t . (a) $\mathcal{M}_{\text{safe} \rightarrow \text{rel}}(t)$: critical safe beams on s^c mapped to helper release beams on s^N/s^S (SAHR). (b) $\mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)$: critical closed beams mapped to helper relay beams (BPLR).

If candidate helper user v is reassociated to b_c^{safe} , the reduction in required power on the helper release beam is

$$\Delta P_v^{\text{rel}}(t) = P_{b_h^{\text{rel}}}^{\text{req,old}}(t) - P_{b_h^{\text{rel}}}^{\text{req,new}}(v, t).$$

The SAHR score is

$$S_{\text{relief}}(v, t) = \Delta P_v^{\text{rel}}(t).$$

Candidates are sorted in descending order of $S_{\text{relief}}(v, t)$. A trial reassociation is accepted only if the transferred user does not lose satisfaction and no original user on the safe beam loses satisfaction:

$$\begin{aligned} \text{sat}_v^{\text{safe}}(t) &\geq \text{sat}_v^{\text{old}}(t), \\ \text{sat}_w^{\text{new}}(t) &\geq \text{sat}_w^{\text{old}}(t), \quad \forall w \in \mathcal{U}_{b_c^{\text{safe}}}^{\text{ori}}(t). \end{aligned}$$

Accepted transfers return $\Delta P_v^{\text{rel}}(t)$ to the helper power pool. For helper satellite s_h , total released power is

$$P_{s_h}^{\text{rel}}(t) = \sum_{v \in \mathcal{A}_{\text{rel}}(s_h, t)} \Delta P_v^{\text{rel}}(t),$$

where $\mathcal{A}_{\text{rel}}(s_h, t)$ is the set of helper users successfully taken over by safe beams.

Unused power under baseline operation is

$$P_{s_h}^{\text{unused}}(t) = P_{s_h}^{\text{max}} - \sum_{b \in \mathcal{B}_{s_h}} P^{\text{base}},$$

so the relay boosting pool is

$$P_{s_h}^{\text{pool}}(t) = P_{s_h}^{\text{unused}}(t) + P_{s_h}^{\text{rel}}(t).$$

All helper beams except release and relay beams remain at baseline power P^{base} .

D. Helper Relay Beam Boosting

After SAHR, helper satellite s_h uses $P_{s_h}^{\text{pool}}(t)$ to boost helper relay beams. The relay beam set is

$$\begin{aligned} \mathcal{B}_{s_h}^{\text{relay}}(t) &= \{b_h^{\text{relay}} : \exists b_c^{\text{off}} \in \mathcal{B}_c^{\text{off}}(t) : \\ &\quad (b_c^{\text{off}}, b_h^{\text{relay}}) \in \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t)\}. \end{aligned}$$

Extra power needed to raise a relay beam from baseline to its upper bound is

$$\Delta P_{b_h^{\text{relay}}}^{\text{max}}(t) = P_{s_h, b_h^{\text{relay}}}^{\text{max}} - P^{\text{base}}.$$

Relay demand for each helper relay beam is

$$D_{b_h^{\text{relay}}}^{\text{relay}}(t) = |\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)|,$$

where $\mathcal{U}_{\text{cand}}(b_h^{\text{relay}}, t)$ is the set of critical users in the mapped closed beam that can be covered by b_h^{relay} .

If $P_{s_h}^{\text{pool}}(t)$ suffices to boost all relay beams to their upper bounds, every relay beam is fully boosted. Otherwise, we apply relay-demand-priority boosting: relay beams are sorted by $D_{b_h^{\text{relay}}}^{\text{relay}}(t)$ in descending order, and higher-demand beams receive power first.

If the remaining pool can raise the current beam to $P_{s_h, b_h^{\text{relay}}}^{\text{max}}$, it is set to the upper bound. If the pool cannot fully boost the next beam but is still positive, the remainder is assigned as a partial boost above P^{base} , avoiding wasted pool power and increasing available relay capacity.

All helper relay beams then proceed to BPLR with final transmit powers that determine their relay capacities.

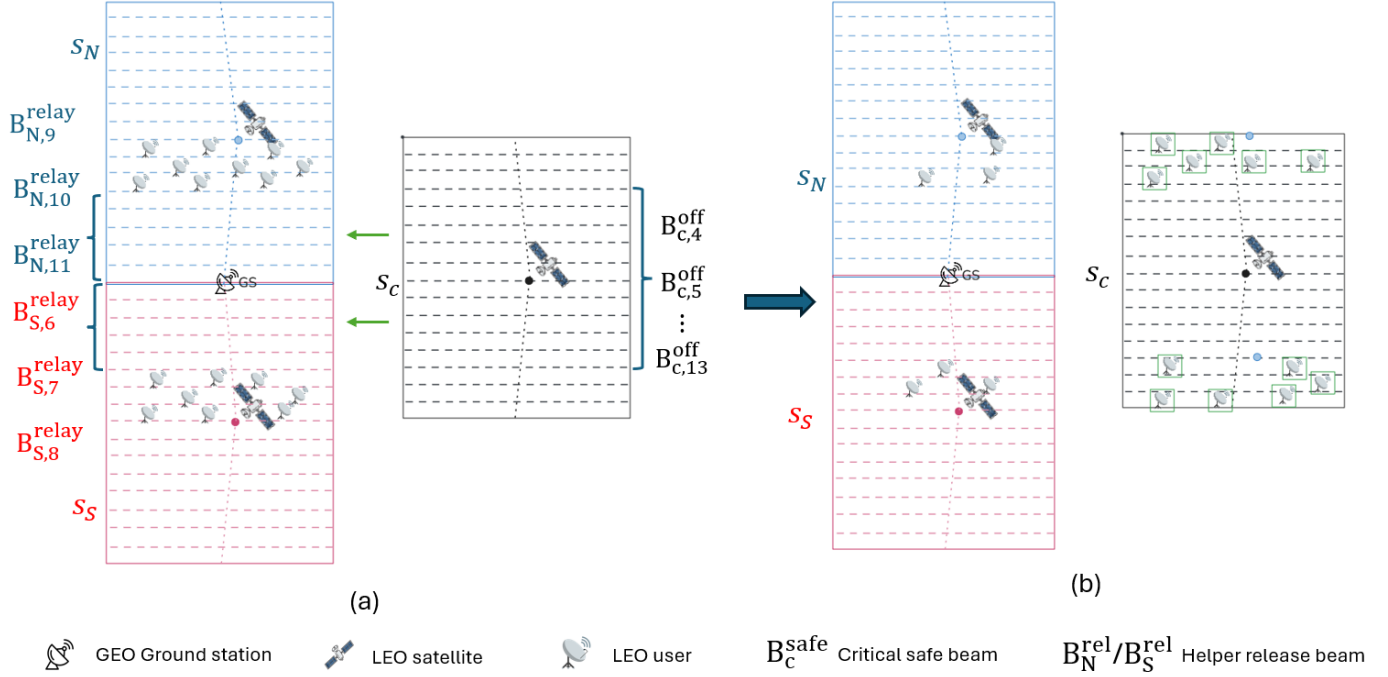


Fig. 5. Illustration of Safe-Beam Assisted Helper Relief (SAHR). Critical safe beams take over eligible helper users from helper release beams, and the released power is returned to the helper satellite power pool.

E. Beam-Pair Local Relay

After relay-beam boosting, BPLR runs within each closed-beam-to-helper-relay-beam pair. Figure 6 illustrates the BPLR process. For

$$(b_c^{\text{off}}, b_h^{\text{relay}}) \in \mathcal{M}_{\text{off} \rightarrow \text{relay}}(t),$$

we identify critical users served by b_c^{off} that also lie in the footprint of b_h^{relay} :

$$\mathcal{U}_{\text{pair}}(b_c^{\text{off}}, b_h^{\text{relay}}, t) = \left\{ u : u \in \mathcal{U}_{s^c, b_c^{\text{off}}}(t), u \in \mathcal{F}_{s_h, b_h^{\text{relay}}}(t) \right\}.$$

The expected achievable rate if candidate critical user u is served by the helper relay beam is

$$R_{u, b_h^{\text{relay}}}(t).$$

The BPLR score is

$$S_{\text{relay}}(u, t) = R_{u, b_h^{\text{relay}}}(t).$$

Candidates are sorted in descending order of $S_{\text{relay}}(u, t)$. The helper relay beam's remaining relay capacity (after protecting existing helper users) is computed from its final transmit power.

Remaining capacity is allocated sequentially to sorted users. If capacity covers demand $D_u(t)$, u is reassigned to the helper relay beam and satisfaction is updated. If the next user cannot be fully served but capacity remains positive, partial capacity is assigned and the pair process stops. When capacity is exhausted, the pair relay process ends.

All relay decisions respect the single-association constraint: a relayed user moves from the critical closed beam to the helper relay beam and is served by only one beam in the time slot.

F. Overall Algorithm

The online procedure uses a single time-slot-based algorithm. In each time slot, SAPR-R first runs SAHR and relay-beam boosting, then runs BPLR using the updated helper relay-beam powers. The complete procedure is given in Algorithm 1.

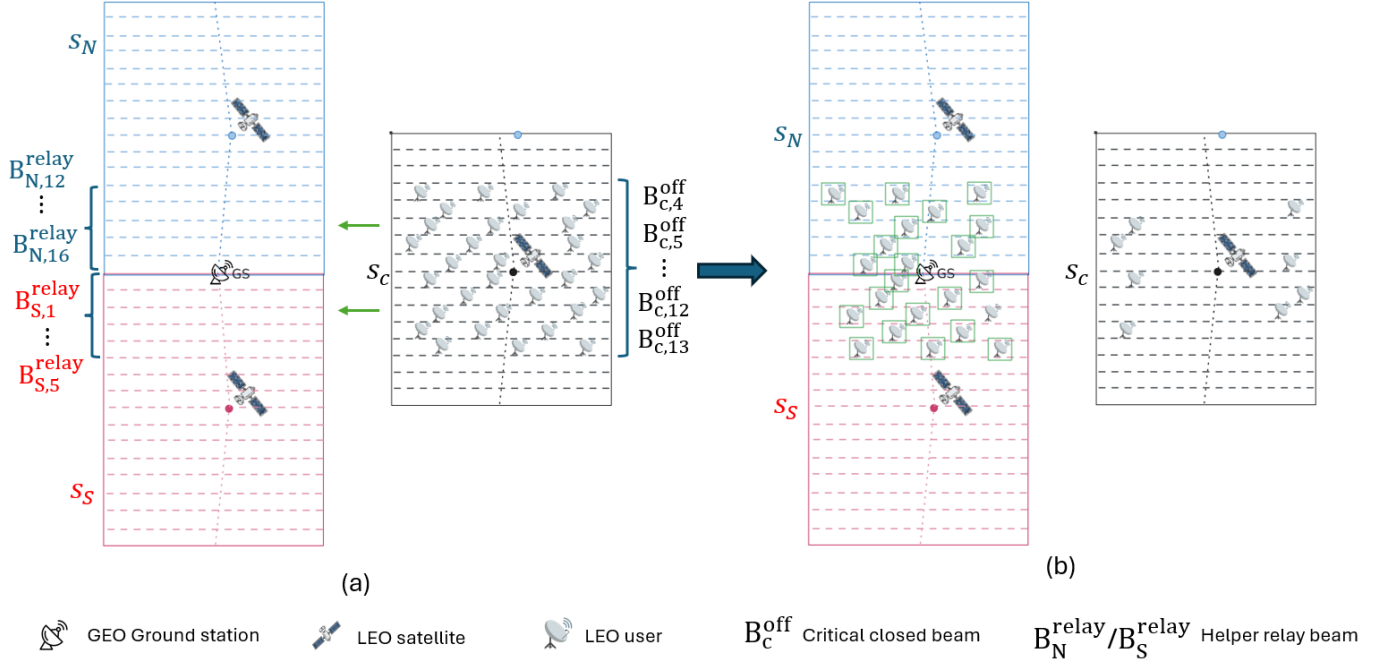


Fig. 6. Illustration of Beam-Pair Local Relay (BPLR). Candidate critical users in closed beams are reassociated to the corresponding helper relay beams according to $\mathcal{M}_{off \rightarrow relay}(t)$.

Algorithm 1 SAPR-R Online Operation

Input: $\{\mathcal{R}_B(t)\}_{t=1}^T$, initial $\mathbf{x}(t)$, $\mathcal{U}(t)$, $D_u(t)$

Input: $P_{s,b}^{base}$, $P_{s,b}^{max}$, P_s^{max} , $F_{s,b}(t)$, capacity/satisfaction model

Output: $\mathbf{x}^{final}(t)$, $P_{s,b}^{final}(t)$, $\text{sat}_u^{final}(t)$

```

1: for  $t = 1$  to  $T$  do
2:   Load  $\mathcal{R}_B(t)$ ; initialize beam powers per record and baseline operation
3:   Stage 1: SAHR and relay-beam boosting
4:   for all  $(b_c^{safe}, b_h^{rel}) \in \mathcal{M}_{safe \rightarrow rel}(t)$  do
5:     Build candidate helper users; compute  $\Delta P_v^{rel}(t)$ 
6:     Sort candidates by  $\Delta P_v^{rel}(t)$  descending
7:     for all  $v$  in sorted order do
8:       Trial reassociation  $v: b_h^{rel} \rightarrow b_c^{safe}$ 
9:       if  $v$  not degraded and users on  $b_c^{safe}$  not degraded then
10:        Accept; update association and  $P_{sh}^{pool}(t)$ 
11:       end if
12:     end for
13:   end for
14:   for all  $s_h \in \{s^N(t), s^S(t)\}$  do
15:     Extract  $\mathcal{B}_{sh}^{relay}(t)$ ; compute relay demand; sort by demand
16:     for all  $b_h^{relay} \in \mathcal{B}_{sh}^{relay}(t)$  in sorted order do
17:       Allocate  $P_{sh}^{pool}(t)$  (full boost, partial boost, or baseline)
18:     end for
19:     Check  $\sum_b P_{s,b}(t) \leq P_s^{max}$ 
20:   end for
21:   Stage 2: Beam-pair local relay
22:   for all  $(b_c^{off}, b_h^{relay}) \in \mathcal{M}_{off \rightarrow relay}(t)$  do
23:     Build candidate critical users; sort by relay link quality
24:     Compute remaining relay capacity of  $b_h^{relay}$  (protect helper users)
25:     for all  $u$  in sorted order do
26:       if capacity  $\geq D_u(t)$  then
27:         Reassociate  $u$  to  $b_h^{relay}$ ; update  $\text{sat}_u(t)$  and capacity
28:       else if remaining capacity  $> 0$  then
29:         Partial allocation to  $u$ ; set remaining capacity to 0; break
30:       else
31:         break
32:       end if
33:     end for
34:   end for
35:   Validate single association, beam power, satellite total power, and EPFD constraints
36:   Store  $\mathbf{x}^{final}(t)$ ,  $P_{s,b}^{final}(t)$ ,  $\text{sat}_u^{final}(t)$ 
37: end for
  
```

G. Complexity Discussion

The proposed method avoids solving the mixed-integer nonlinear program directly. Because $\mathcal{R}_B(t)$ is precomputed, the online procedure does not re-search critical/helpers or mappings; it performs local reassociation and relay only within predefined beam pairs.

Let K_t^{rel} and K_t^{relay} denote the numbers of SAHR helper candidates and BPLR critical candidates in time slot t . In Algorithm 1, building candidates and computing released power are linear in K_t^{rel} , and sorting dominates:

$$\mathcal{O}(K_t^{rel} \log K_t^{rel}).$$

In the BPLR stage, sorting by relay link quality and sequential allocation yield

$$\mathcal{O}(K_t^{relay} \log K_t^{relay}).$$

Over all time slots,

$$\mathcal{O}\left(\sum_{t=1}^T K_t^{rel} \log K_t^{rel} + K_t^{relay} \log K_t^{relay}\right).$$

Online decision is thus limited to local sorting and sequential allocation within precomputed beam pairs, rather than global association and continuous power optimization every time slot.

H. Summary

This chapter presented SAPR-R to address the tradeoff between EPFD compliance and user satisfaction when the critical satellite approaches the protected GS. We first build time-slot beam relation records from predictable geometry and footprint overlap, then run SAHR, relay-beam boosting, and BPLR online.

The staged design decouples user association and beam power allocation into local steps: critical safe beams release helper power; helper relay beams receive extra transmit power by relay demand; and closed-beam users are served within beam pairs. The method improves total user satisfaction during critical periods while satisfying the EPFD constraint.

V. EVALUATION

A. Evaluation Overview

This chapter evaluates the performance of the proposed Safe-Beam Assisted Power-Release Relay (SAPR-R) method, focusing on its ability to improve user satisfaction and service continuity under the aggregate EPFD constraint. The evaluation consists of three main parts. First, we observe the aggregate EPFD variation during the critical satellite pass over the GS, as well as the number of beams that must be shut down to satisfy the EPFD constraint. This analysis confirms the existence of EPFD violation and beam shutdown demand during the critical pass, and further supports the rationale of designing a relay-based recovery method. Second, we compare different interference mitigation methods in terms of service performance and relay capability. Finally, we further analyze the impact of different EPFD limits on user satisfaction to evaluate the stability and applicability of SAPR-R under different interference constraint conditions.

B. Simulation Setup

To evaluate the performance of the proposed SAPR-R method under the NGSO–GSO coexistence scenario, we design a simulation framework that combines STK and MATLAB. This framework is used to construct the LEO constellation, obtain the time-varying satellite geometry, and further evaluate the performance of different interference mitigation methods in terms of EPFD compliance, downlink capacity, and user satisfaction.

The simulation uses the following two main tools:

- **STK (Satellite Tool Kit):** [19] STK is used to construct the OneWeb-like LEO constellation and generate the relative positions, visibility relationships, and beam coverage information among the LEO satellites, the GS, and users at each time slot. Based on the system information generated by STK, the dynamic geometric variation during the critical satellite pass near the GS can be simulated.
- **MATLAB:** [20] MATLAB is used to process the system information generated by STK and perform offline preprocessing. This includes constructing the beam–user association, critical/helper satellite relationship, and critical-beam/helper-beam mappings. MATLAB is then used to execute the proposed SAPR-R method and calculate the EPFD and user satisfaction performance of each method at different time slots.

The main simulation parameters used in this chapter are summarized in Table I. The table integrates the settings related to the LEO constellation, GSO satellite, GS, LEO user ground station, and EPFD evaluation. These parameters are used in the following EPFD evaluation and user satisfaction comparison.

C. Compared Schemes

To evaluate the proposed SAPR-R method, we compare it with several interference mitigation methods. All methods are evaluated under the same simulation parameters to ensure a fair comparison. The compared schemes are described as follows:

- **Beam shutdown only:** When the aggregate EPFD exceeds the limit, the high-interference beams on the critical satellite are directly shut down to satisfy the EPFD constraint. This method does not perform relay.
- **PC + Tilt:** This method represents a power-control and satellite-tilting based interference mitigation strategy. It reduces the interference toward the GS by adjusting the transmit power and satellite tilt angle. Although this method can help satisfy the EPFD constraint, it does not provide a relay-based recovery mechanism for users.
- **Only BPLR:** This method only applies the BPLR stage. Helper relay beams serve users located under the critical closed beams using the default power of the helper satellite. The SAHR stage is not applied.
- **SAPR-R (Proposed):** This method includes both the SAHR and BPLR stages.

D. Evaluation Results

Based on the simulation setup and compared schemes described above, this section evaluates the performance of SAPR-R during the critical pass. The analysis is divided into four parts. First, the aggregate EPFD and the number of closed beams are examined to verify whether SAPR-R can still satisfy the EPFD constraint after relay-based recovery, and to observe the beam shutdown behavior of the critical satellite at different time slots. Second, the average satisfaction of different methods is compared under different user loads to analyze how increasing system load affects service performance. Third, the number of relayed critical users is used to explain the main reason why SAPR-R improves satisfaction, and the satisfaction CDF is further used to observe the satisfaction distribution of all users. Finally, the average user satisfaction of SAPR-R under different EPFD limits is evaluated to observe how the strictness of the interference limit affects service performance.

1) EPFD Compliance and Beam Shutdown:

This subsection first analyzes the aggregate EPFD after applying SAPR-R and further observes the number of beams that must be shut down on the critical satellite to satisfy the EPFD constraint. This analysis is used to verify whether the proposed method can still satisfy the EPFD limit after relay-based recovery, and to show the beam shutdown behavior during the critical pass.

Fig. 7 shows the aggregate EPFD variation during the critical satellite pass near the GS. The red line represents the EPFD constraint, and the black line represents the aggregate EPFD after applying SAPR-R. As shown in the figure, SAPR-R keeps the aggregate EPFD below the limit during the entire observation period. This indicates that the proposed method does not cause EPFD violation while performing service recovery for closed-beam users. Therefore, SAPR-R can compensate for the service loss caused by beam shutdown while maintaining EPFD compliance.

TABLE I
SIMULATION PARAMETERS.

Parameter	Value
LEO constellation	
Altitude (km)	1200
Orbit plane inclination	87.9°
Time slot	1 s
Maximum transmit power	16.8 W per satellite
Maximum antenna gain	34.33 dBi
Downlink frequency	11.7 GHz
Bandwidth	250 MHz
Antenna pattern	[14] $A \exp(\beta\phi)$
Maximum tilt angle	10°
Approximated antenna gain coefficient, (A)	4.0×10^3
Approximated antenna gain coefficient, (B)	-0.0671
GSO satellite	
Altitude (km)	35785.4
Longitude	30.6
Latitude	0
Downlink frequency	11.7 GHz
GSO ground station	
Minimum elevation angle	10°
Maximum receive gain	38.1 dBi
Noise temperature	240 K
Antenna pattern	[8]
LEO user ground station	
Maximum receive gain	40.95 dBi
Noise temperature	240 K
Antenna pattern	[8]
User demand	50 Mbps
EPFD	
EPFD limit (dB(W/m ² /1 MHz))	-173.4 [5]
Reference antenna diameter	60 cm

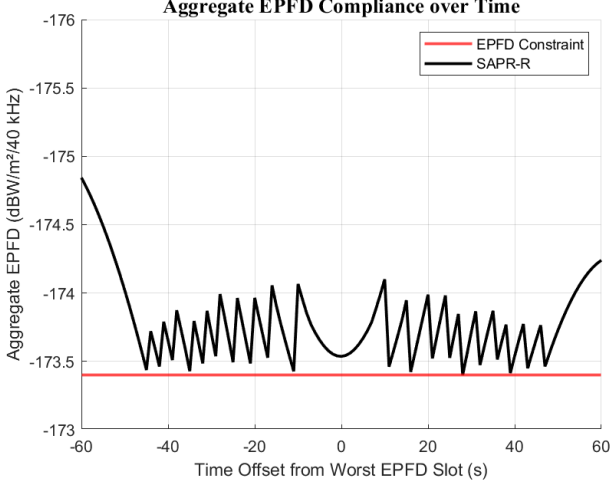


Fig. 7. Aggregate EPFD over time during the critical pass.

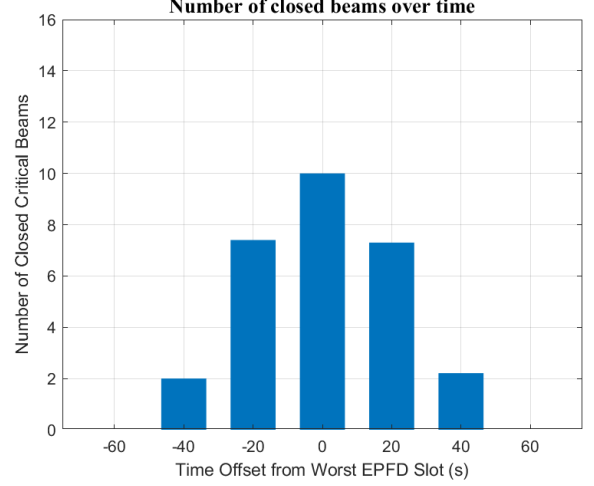


Fig. 8. Number of closed beams over time during the critical pass.

Fig. 8 shows the number of beams that must be shut down on the critical satellite at different time offsets. When the satellite approaches the worst EPFD slot, the number of closed beams gradually increases and reaches its maximum around the worst EPFD slot. After the satellite moves away from GS, the number of closed beams gradually decreases. This result shows that the beam shutdown demand changes with the satellite-GS geometry.

2) User Satisfaction under Different User Loads:

This subsection compares the average satisfaction of different

interference mitigation methods under different user loads. By changing the number of users, we can observe how the service performance changes as the system load increases, and evaluate whether SAPR-R can maintain user satisfaction under different load conditions.

Fig. 9 shows the average satisfaction under three user load scenarios: 30 users, 50 users, and 70 users. As shown in the figure, Beam shutdown only has the lowest satisfaction because it does not provide relay service for users affected by closed beams. PC + Tilt can reduce the interference toward the GS,

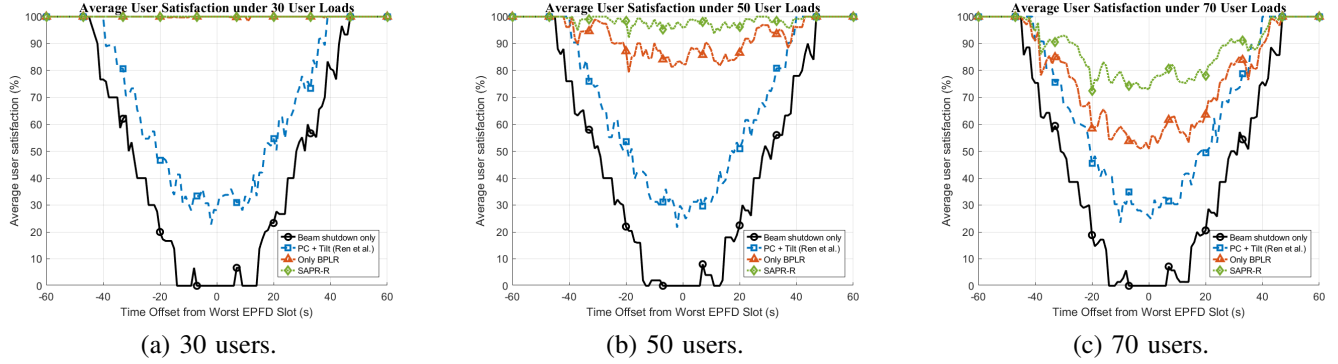


Fig. 9. Average satisfaction under different user loads. (a) 30 users. (b) 50 users. (c) 70 users.

but it does not recover the service of users affected by beam shutdown. Only BPLR can serve part of the closed-beam users through helper relay beams, so it achieves better satisfaction than Beam shutdown only and PC + Tilt.

Compared with the other methods, SAPR-R maintains higher average satisfaction under all three user loads. When the user load is low, the helper satellite usually has enough available power, so the difference between Only BPLR and SAPR-R is small. However, as the user load increases, more closed-beam users need relay service, and Only BPLR becomes limited by the default power of the helper satellite. In contrast, SAPR-R uses SAHR to release available power from the helper satellite and improve the relay capability of helper relay beams. Therefore, SAPR-R can maintain more stable service performance under medium and high user loads.

3) Relay Capability and Satisfaction Distribution:

This subsection further analyzes why SAPR-R improves user satisfaction and observes the satisfaction distribution of different methods. Since 30 users is a low-load scenario, the default available power of the helper satellite is usually sufficient to support most relay demand. Therefore, the difference in relay capability is not obvious under 30 users. To better show the effect of the proposed algorithm, this subsection focuses on the 50-user and 70-user scenarios.

Fig. 10 compares the number of successfully relayed critical users between Only BPLR and SAPR-R under the 50-user and 70-user scenarios. As shown in the figure, SAPR-R serves more critical users in both load scenarios. The difference becomes more obvious around the worst EPFD slot, where the beam shutdown demand is the most severe. This result shows the necessity of Stage 1 in SAPR-R.

Fig. 11 shows the satisfaction CDF of three interference mitigation methods. This figure focuses on the user-level satisfaction distribution. A CDF curve that shifts further to the right means that more users maintain higher satisfaction. In contrast, if the curve rises quickly in the low-satisfaction region, more users suffer from service degradation.

As shown in the figure, the CDF curve of SAPR-R shifts more toward the high-satisfaction region than the other methods. This means that more users can maintain better service quality. The result shows that SAPR-R not only improves average satisfaction, but also reduces the proportion of low-satisfaction users, making the overall user service quality more stable.

4) Impact of Different EPFD Limits:

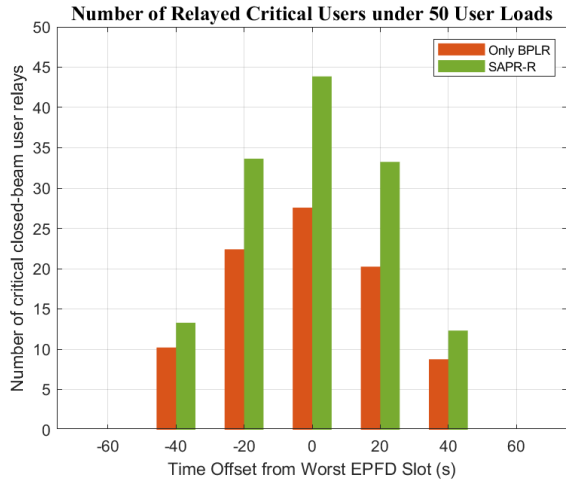
This subsection further evaluates the impact of different EPFD limits on the service performance of SAPR-R. When the EPFD limit becomes stricter, the allowed aggregate interference becomes lower. As a result, the critical satellite may need to shut down more beams to satisfy the constraint, and the relay demand of closed-beam users may also increase. This analysis is used to observe whether SAPR-R can maintain stable average user satisfaction under different EPFD limits.

Fig. 12 shows the average user satisfaction of SAPR-R under different EPFD limits. As shown in the figure, when the EPFD limit is more relaxed, SAPR-R can maintain higher average user satisfaction. This result shows that the strictness of the EPFD limit directly affects user satisfaction. It also indicates that SAPR-R can achieve better service performance under more relaxed interference constraints.

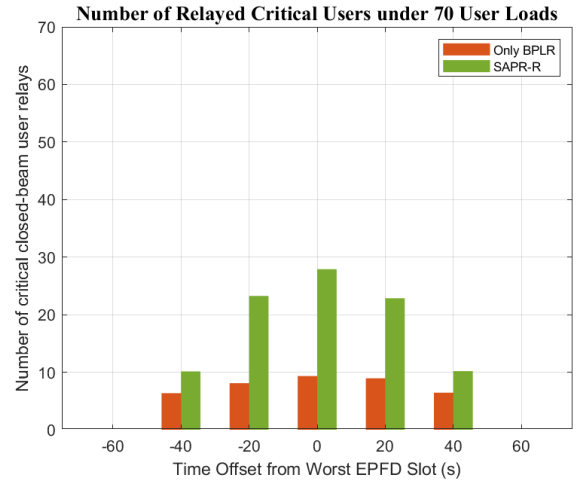
Overall, SAPR-R keeps the aggregate EPFD below the limit during the critical pass and uses SAHR to improve the ability of helper relay beams to serve critical users. Compared with other interference mitigation methods, SAPR-R maintains higher average satisfaction under different user loads and reduces the proportion of low-satisfaction users. In addition, the results under different EPFD limits show that the strictness of the interference constraint affects user satisfaction, and SAPR-R can maintain better service performance under more relaxed constraints. Overall, SAPR-R achieves a better balance between EPFD compliance and service continuity.

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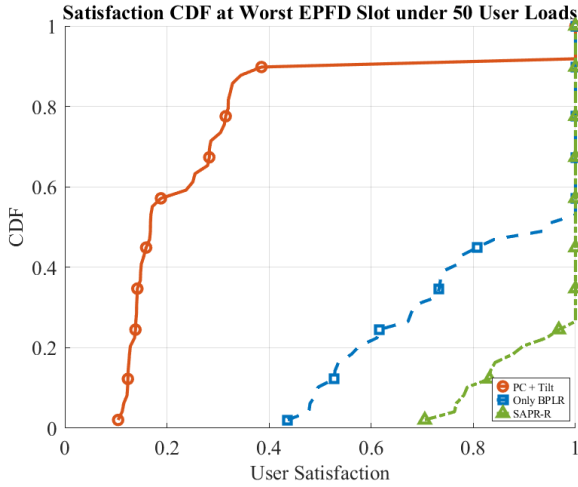


(a) 50 users.

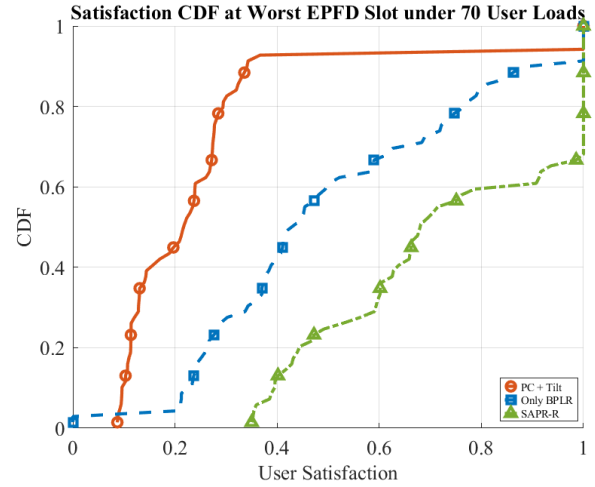


(b) 70 users.

Fig. 10. Number of relayed critical users with and without SAHR. (a) 50 users. (b) 70 users.



(a) 50 users.



(b) 70 users.

Fig. 11. CDF of user satisfaction under different interference mitigation methods.

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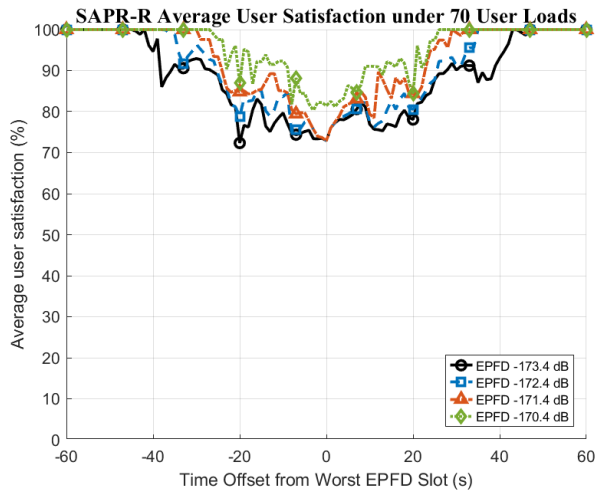


Fig. 12. Average user satisfaction of SAPR-R under different EPFD limits.

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